

Structural Description of the Collatz Inverse Tree via Mersenne-Fermat Factorization and Jacobsthal Sequences

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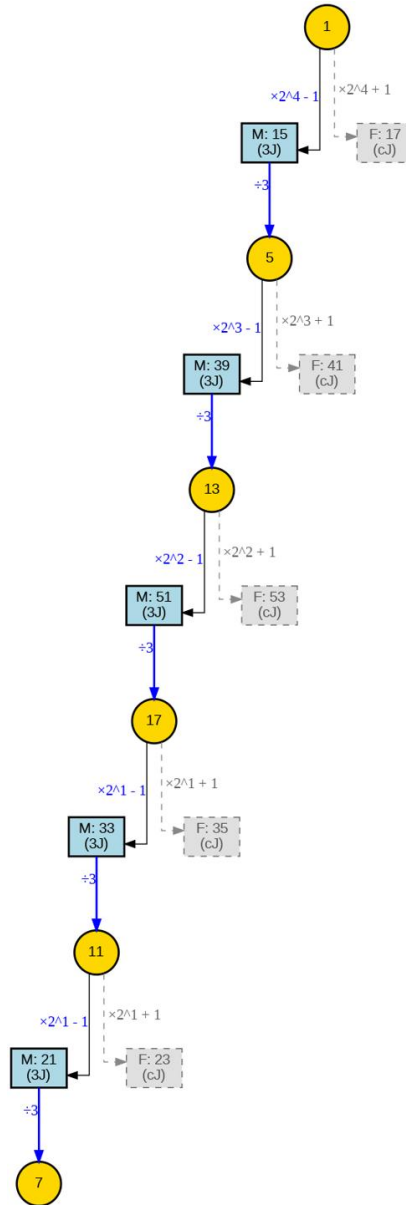


Figure 1. Mersenne-Fermat cascade decay of odd nuclei.

The diagram illustrates the reverse odd Collatz orbit for the initial value 7 ($1 \rightarrow 5 \rightarrow 13 \rightarrow 17 \rightarrow 11 \rightarrow 7$), demonstrating the separation of M-type (3J) and F-type (cJ) factors at each generation.

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